

VATTIKUTI MOHIT

AI/ML Engineer | Full-Stack Developer | Final-Year CSE-AI Student

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🔗 HackerRank

Summary

Final-year CSE-AI student who ships working systems, not just prototypes – a CNN-based audio deepfake detector (121,000+ samples, 23.93% EER) and a userspace TCP/IP stack built from raw bytes in C++. Cut model size by ~90% and defect turnaround time by 35%. Seeking an entry-level engineering role.

Education

Pragati Engineering College, Andhra Pradesh *B.Tech in CSE – Artificial Intelligence* *Sept 2023 – Present*
GPA: 7.2/10

Technical Skills

Languages: C++, Python, Java, JavaScript, SQL, C

AI/ML: PyTorch, TensorFlow, Scikit-learn, Librosa, MediaPipe, Grad-CAM, Computer Vision

Web & Systems: React, Node.js, HTML/CSS, REST APIs, TCP/IP, ICMP, TUN/TAP, CMake

Tools: Git/GitHub, Streamlit, Hugging Face, Power BI, AWS, Microsoft SQL Server

Projects

Userspace TCP/IP Stack (tinystack)

[demo](#) [code](#)

- Built a complete TCP/IP stack in C++ running entirely in userspace via macOS utun, with zero OS networking calls and zero external libraries – raw bytes only.
- Implemented the full RFC 793 TCP state machine (LISTEN → SYN_RECEIVED → ESTABLISHED → FIN → CLOSED), reaching sub-millisecond ICMP round-trips (0.21 ms average, 0% loss over 40 packets).
- Designed an internal packet router supporting both a TCP server (HTTP/1.0, port 8080) and an outbound TCP client, enabling full HTTP request/response cycles outside the kernel network stack.
- **Tech Stack:** C++20, CMake, ICMP, TCP, RFC 793, TUN/TAP, non-blocking I/O, RFC 1071 checksum

Audio Deepfake Detector

[demo](#) [code](#)

- Trained a CNN-based spoofed-speech classifier on the ASVspoof 2019 LA dataset (121,000+ samples), reaching a 23.93% Equal Error Rate on attack types withheld from training.
- Built a Mel-spectrogram pipeline (Librosa) feeding a 4-block CNN with Batch Normalisation and Global Average Pooling, cutting parameter count by ~90% while correcting a 9:1 class imbalance via weighted sampling.
- Fixed a data-leakage bug via systematic audio trimming, added Grad-CAM explainability, and deployed an interactive Streamlit demo on Hugging Face Spaces.
- **Tech Stack:** Python, PyTorch, Librosa, NumPy, Pandas, Scikit-learn, Streamlit

NeonAura AI – XR Hand-Tracking Platform

[demo](#) [code](#)

- Shipped a zero-backend, client-side XR app (Three.js, MediaPipe Hands, WebGL) turning a standard webcam into a gesture-controlled motion platform with 10+ distinct interactions.
- Built an Air Drawing Studio with live letter recognition and a shader-based CGI Liquid Hand with physics-driven finger deformation; cut frame latency by 30%+ across all gestures.
- Released six visual themes (Rainbow, Cyberpunk, Lava, Ocean, Galaxy, Plasma) and an AR mode with real-time hand-skeleton tracking – no server dependencies.
- **Tech Stack:** JavaScript (ES6), Three.js, MediaPipe Hands, WebGL, Canvas API

Achievements

- Earned a 5-star HackerRank rating across 30+ SQL challenges, speeding up complex query execution by 40% through indexing and rewrites; cut DSA time complexity from $O(n^2)$ to $O(n \log n)$, lifting runtime performance by 30%.
- Co-built 2 full-stack applications (JavaScript, React, SQL); reduced defect turnaround time by 35% through modular architecture and code reviews.

Certifications

Google UX Design Certificate • Deloitte Data Analytics Simulation • AWS Solutions Architecture Simulation • Python & Generative AI – Infosys SpringBoard • edX Computer Science 101